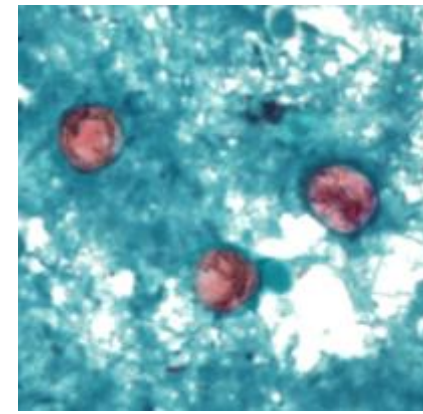
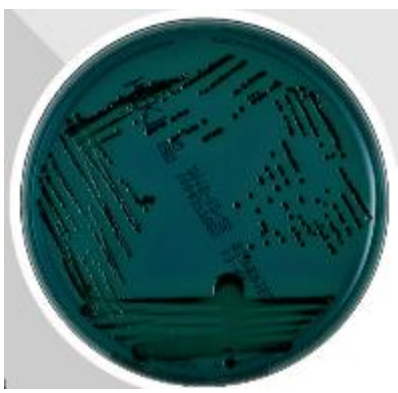
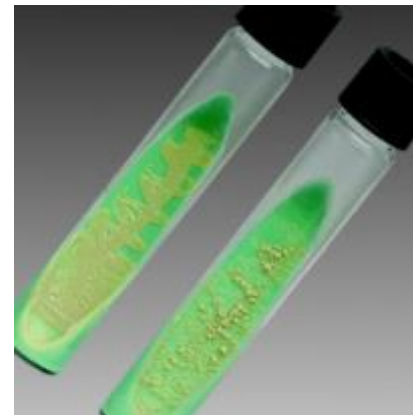
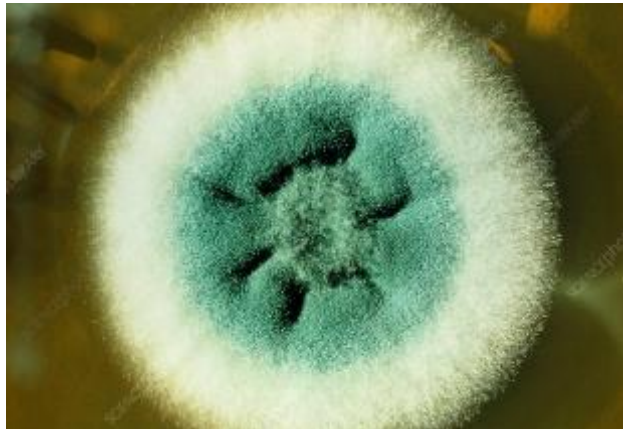
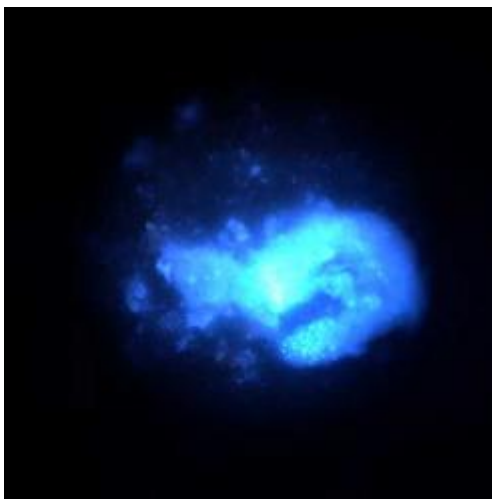
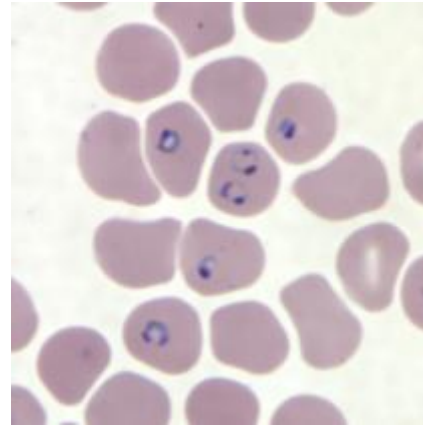




Introduction to the Microbiology Laboratory

Ella Martin, MD

July 7, 2025



We Diagnose Infections!

- Sepsis
- Pneumonia
- Urinary tract infections
- Wounds
- Meningitis
- Infectious diarrhea
- Sexually-transmitted infections

We Diagnose Infections!

Bacteriology

Mycology

Parasitology

Virology

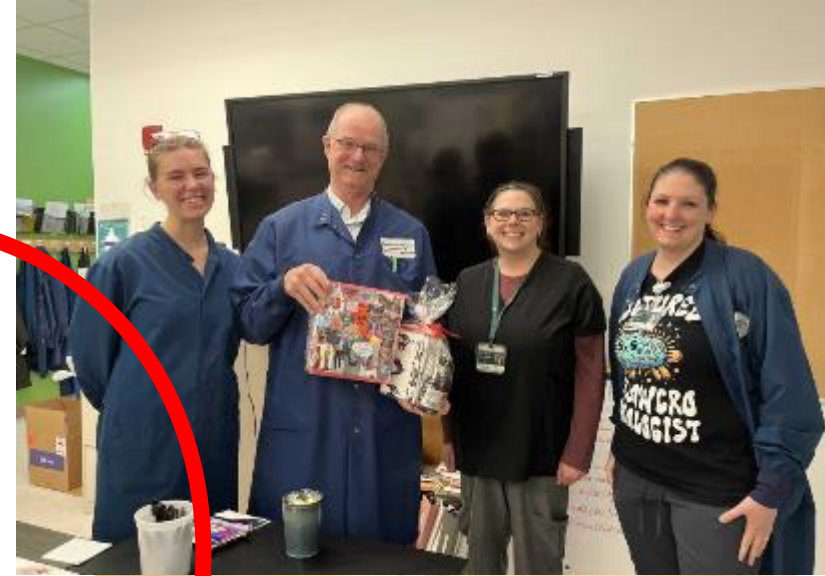
Mycobacteriology

We Diagnose Infections!

- ~350,000 tests per year
- Open 24/7/365

Why I Love My Job

Intellectually stimulating
Team-based
Ever-changing
Forest vs. trees
Local vs. national



What do we do all day?

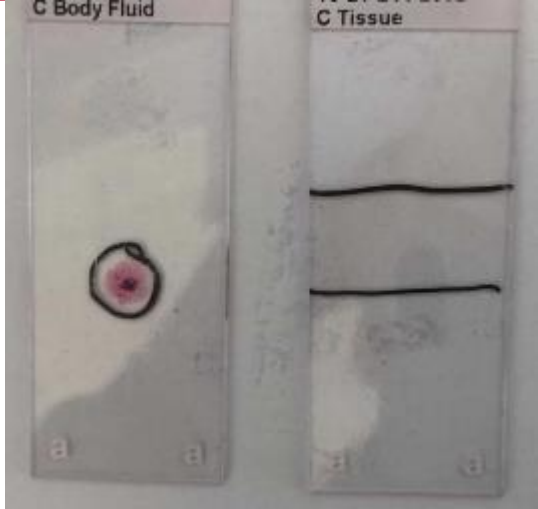
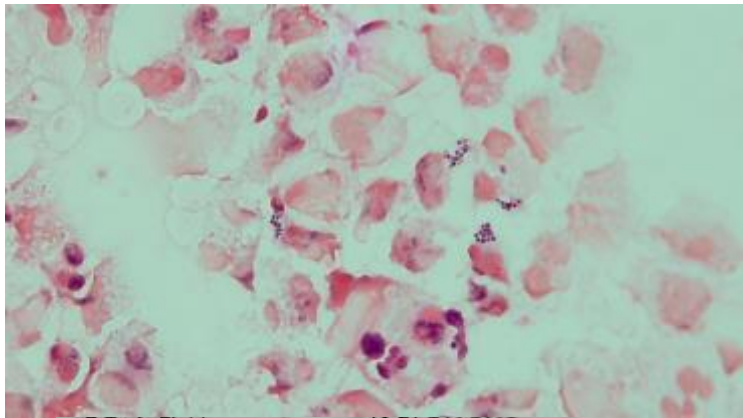
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Patient Sample



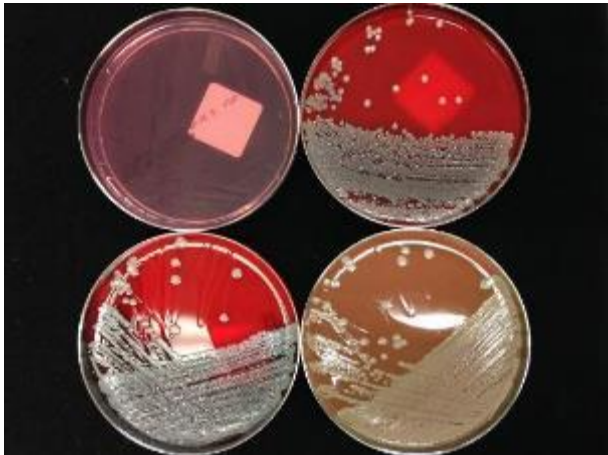
2

Gram stain



3

Grow pathogen



4

Test antibiotics

Susceptibility

	Staphylococcus aureus VITEK 2 METHOD
Cefazolin	Sensitive
Ceftriaxone	Sensitive
Ciprofloxacin	Sensitive
Clindamycin	Sensitive
Erythromycin	Resistant
Gentamicin	Sensitive ¹
Levofloxacin	Sensitive
Oxacillin	Sensitive ²
Tetracycline	Sensitive
Trimethoprim/Sulfa	Sensitive
Vancomycin	Sensitive

Shout-out to the Human Microbiome



Our Natural Defenses, Part 1

- Skin
 - Dense, keratinized layer
 - Low pH (<5.5)
 - Fatty acids
- GI Tract
 - Acidic gastric secretions
 - Bile & pancreatic enzymes
 - Normal protective microbiota
 - Layer of protective mucus
 - Mucosal antimicrobial peptides (defensins)
 - Secreted IgA antibodies

Our Natural Defenses, Part 2

- Respiratory Tract
 - Mucociliary clearance
 - Alveolar macrophages
- Urogenital tract
 - Urination
 - Normal vaginal microbiota
 - Epithelium

Concepts of Pathogenesis

- Colonization
- Invasion
- Virulence
- Cellular Responses

